

PHASE 3 MODEL EVALUATION REPORT

Claim Outcome Classification

Model B — Multiclass Classification (Paid / Pending / Rejected)

1. Executive Summary

This report presents the training and testing evaluation metrics for all four machine learning models developed in Phase 3 to classify hospital insurance claim outcomes: Paid (0), Pending (1), and Rejected (2). Four algorithms were benchmarked: Logistic Regression, Random Forest, HistGradientBoosting, and XGBoost. HistGradientBoosting achieved the highest test accuracy (50.70%) and was selected as the production model.

All models show notable improvement over a random 33% baseline, with ensemble methods reaching 46-51% accuracy. The Rejected class is well-captured by tree-based models with recall of 60-74%, which is a critical operational strength. The Approved class is predicted with high precision (0.74-0.78) but moderate recall.

Model Performance at a Glance

Model	Train Acc.	Test Acc.	Macro F1	Weighted F1
Logistic Regression	34.74%	35.20%	0.32	0.38
Random Forest	50.79%	46.12%	0.41	0.48
HistGradientBoosting (Saved)	54.43%	50.70%	0.45	0.53
XGBoost	46.75%	45.78%	0.41	0.48

2. Dataset & Setup

2.1 Data Overview

The dataset `capstone_phase3_features.csv` contains 25,000 hospital visit records. After dropping ID columns, the model targets `claim_status` as the outcome variable. Label encoding: 0 = Approved, 1 = Pending, 2 = Rejected.

2.2 Feature Selection

Numerical Features (11): `billed_amount`, `bill_per_hour`, `payment_days`, `provider_rejection_rate`, `length_of_stay_hours`, `visit_frequency`, `avg_length_of_stay_patients`, `days_since_registration`, `age`, `chronic_flag`, `is_backdated`

Categorical Features (4, one-hot encoded): `insurance_provider`, `department`, `visit_type`, `gender`

2.3 Class Distribution

Class	Train Count	Test Count
Approved (0)	11,952	2,988
Pending (1)	5,010	1,253
Rejected (2)	3,038	759
Total	20,000	5,000

Note: Significant class imbalance — Approved represents ~60% of data. Balanced sample weights applied: Approved=0.558, Pending=1.331, Rejected=2.194.

3. Model 1 — Logistic Regression

Configuration

Solver: lbfgs | Max Iterations: 3,000 | Class Weight: Balanced | Random State: 42

3.1 Training Metrics

Training Accuracy: 34.74%

Class	Precision	Recall	F1-Score	Support
Approved (0)	0.59	0.35	0.44	11,952
Pending (1)	0.29	0.36	0.32	5,010
Rejected (2)	0.14	0.30	0.19	3,038
Macro Avg	0.34	0.34	0.32	20,000
Weighted Avg	0.44	0.35	0.37	20,000

Training Confusion Matrix

Actual \ Predicted	Paid	Pending	Rejected
Approved	6,295	641	5,016
Pending	1,404	1,299	2,307
Rejected	360	115	2,563

3.2 Testing Metrics

Test Accuracy: 35.20%

Class	Precision	Recall	F1-Score	Support
Approved (0)	0.61	0.37	0.46	2,988
Pending (1)	0.29	0.35	0.32	1,253
Rejected (2)	0.13	0.28	0.18	759
Macro Avg	0.34	0.33	0.32	5,000
Weighted Avg	0.45	0.35	0.38	5,000

Test Confusion Matrix

Actual \ Predicted	Paid	Pending	Rejected
Approved	1,108	813	1,067

Actual \ Predicted	Paid	Pending	Rejected
Pending	452	440	361
Rejected	268	279	212

Observation: Logistic Regression is the weakest model at 35.20% test accuracy, barely above random. Very low Rejected recall (28%) and precision (13%) make it unsuitable for claim outcome classification. Near-identical train/test accuracy confirms underfitting.

4. Model 2 — Random Forest

Configuration

n_estimators: 300 | max_depth: 8 | min_samples_split: 8 | min_samples_leaf: 5 | max_features: sqrt
| Class Weight: Balanced | bootstrap: False

4.1 Training Metrics

Training Accuracy: 50.79%

Class	Precision	Recall	F1-Score	Support
Approved (0)	0.78	0.53	0.63	11,952
Pending (1)	0.63	0.26	0.37	5,010
Rejected (2)	0.26	0.84	0.40	3,038
Macro Avg	0.56	0.54	0.46	20,000
Weighted Avg	0.66	0.51	0.53	20,000

Training Confusion Matrix

Actual \ Predicted	Paid	Pending	Rejected
Approved	6,295	641	5,016
Pending	1,404	1,299	2,307
Rejected	360	115	2,563

4.2 Testing Metrics

Test Accuracy: 46.12%

Class	Precision	Recall	F1-Score	Support
Approved (0)	0.75	0.50	0.60	2,988
Pending (1)	0.47	0.19	0.27	1,253
Rejected (2)	0.23	0.74	0.35	759
Macro Avg	0.48	0.48	0.41	5,000
Weighted Avg	0.60	0.46	0.48	5,000

Test Confusion Matrix

Actual \ Predicted	Paid	Pending	Rejected
Approved	1,508	188	1,292

Actual \ Predicted	Paid	Pending	Rejected
Pending	381	236	636
Rejected	119	78	562

4.3 Top Feature Importances

Feature	Importance Score	% of Total
billed_amount	0.3509	35.1%
payment_days	0.3178	31.8%
bill_per_hour	0.1313	13.1%
length_of_stay_hours	0.0440	4.4%
avg_length_of_stay_patients	0.0337	3.4%
days_since_registration	0.0300	3.0%
age	0.0289	2.9%
visit_frequency	0.0162	1.6%
provider_rejection_rate	0.0068	0.7%

Observation: Random Forest achieves 46.12% test accuracy with an impressive Rejected recall of 74%. The 4.67% train-test gap reflects controlled overfitting. Pending recall (19%) is very low — a key weakness. billed_amount and payment_days together account for ~67% of feature importance.

5. Model 3 — HistGradientBoosting (Production Model)

Configuration

learning_rate: 0.01 | max_iter: 400 | max_depth: 5 | l2_regularization: 0.8 | Sample Weights: Balanced

5.1 Training Metrics

Training Accuracy: 54.43%

Class	Precision	Recall	F1-Score	Support
Approved (0)	0.76	0.60	0.67	11,952
Pending (1)	0.49	0.33	0.40	5,010
Rejected (2)	0.28	0.67	0.40	3,038
Macro Avg	0.51	0.53	0.49	20,000
Weighted Avg	0.62	0.54	0.56	20,000

Training Confusion Matrix

Actual \ Predicted	Paid	Pending	Rejected
Approved	7,047	1,481	3,424
Pending	1,554	1,660	1,796
Rejected	671	330	2,037

5.2 Testing Metrics

Test Accuracy: 50.70%

Class	Precision	Recall	F1-Score	Support
Approved (0)	0.74	0.57	0.65	2,988
Pending (1)	0.41	0.30	0.35	1,253
Rejected (2)	0.26	0.60	0.36	759
Macro Avg	0.47	0.49	0.45	5,000
Weighted Avg	0.58	0.51	0.53	5,000

Test Confusion Matrix

Actual \ Predicted	Paid	Pending	Rejected
Approved	1,839	384	765

Actual \ Predicted	Paid	Pending	Rejected
Pending	454	363	436
Rejected	191	127	441

Observation: HistGradientBoosting is the clear production winner — highest test accuracy (50.70%), best Macro F1 (0.45), best Weighted F1 (0.53), and the most balanced performance across all three classes. It achieves 57% Approved recall and 60% Rejected recall — the best overall tradeoff. Saved as claim_model.pkl.

6. Model 4 — XGBoost

Configuration

n_estimators: 250 | learning_rate: 0.01 | max_depth: 2 | subsample: 0.8 | colsample_bytree: 0.5 |
reg_lambda: 1 | Objective: multi:softprob | Sample Weights: Balanced

6.1 Training Metrics

Training Accuracy: 46.75%

Class	Precision	Recall	F1-Score	Support
Approved (0)	0.75	0.49	0.59	11,952
Pending (1)	0.44	0.25	0.32	5,010
Rejected (2)	0.24	0.74	0.36	3,038
Macro Avg	0.48	0.49	0.42	20,000
Weighted Avg	0.60	0.47	0.49	20,000

Training Confusion Matrix

Actual \ Predicted	Paid	Pending	Rejected
Approved	4,228	3,400	4,324
Pending	1,884	1,819	1,307
Rejected	1,078	1,060	900

6.2 Testing Metrics

Test Accuracy: 45.78%

Class	Precision	Recall	F1-Score	Support
Approved (0)	0.75	0.49	0.59	2,988
Pending (1)	0.40	0.23	0.29	1,253
Rejected (2)	0.23	0.71	0.35	759
Macro Avg	0.46	0.48	0.41	5,000
Weighted Avg	0.58	0.46	0.48	5,000

Test Confusion Matrix

Actual \ Predicted	Paid	Pending	Rejected
Approved	1,461	332	1,195

Actual \ Predicted	Paid	Pending	Rejected
Pending	361	286	606
Rejected	125	92	542

Observation: XGBoost achieves 45.78% test accuracy, the lowest among tree-based models. Shallow depth (max_depth=2) limits capacity. Rejected recall (71%) is solid but Pending recall (23%) is the worst across all models. High train-to-test consistency suggests capacity constraint rather than overfitting.

7. Fairness & Subgroup Analysis (Random Forest)

Subgroup analysis was performed using Random Forest predictions on the test set, evaluating Rejected claim recall across demographic and provider groups.

7.1 Rejected Recall by Gender

Gender	Rejected Recall
Female	74%
Male	74%

7.2 Rejected Recall by City

City	Rejected Recall
Hyderabad	80%
Delhi	78%
Mumbai	75%
Chennai	74%
Bangalore	71%
Pune	68%

7.3 Rejected Recall by Insurance Provider

Insurance Provider	Rejected Recall
CareOne	80%
MediCareX	76%
HealthPlus	72%
SecureLife	68%

Key Finding: Rejected recall is equitable across gender (74% for both). City-level variation ranges from 68% (Pune) to 80% (Hyderabad); insurer-level from 68% (SecureLife) to 80% (CareOne). No extreme disparities exist, though Pune patients and SecureLife policyholders are slightly disadvantaged in rejected claim detection.

8. Overall Summary & Recommendations

8.1 Model Ranking

Rank	Model	Test Acc.	Macro F1	Rej. Recall	Status
1	HistGradientBoosting	50.70%	0.45	60%	Production
2	Random Forest	46.12%	0.41	74%	Candidate
3	XGBoost	45.78%	0.41	71%	Benchmarked
4	Logistic Regression	35.20%	0.32	28%	Baseline

8.2 Key Findings

1. HistGradientBoosting is the clear winner — it leads on test accuracy (50.70%), Macro F1 (0.45), and Weighted F1 (0.53), providing the best overall balance across all three claim classes. Correctly selected as the production model.

2. The Rejected class is well-captured by tree-based models (60-74% recall), which is critical for identifying denied insurance claims. This is a major strength of Model B compared to Model A's High-class recall.

3. The Pending class is the hardest to classify across all models (F1: 0.27-0.40). Its intermediate nature and feature overlap with both Approved and Rejected makes precise classification inherently difficult.

4. billed_amount (35.1%) and payment_days (31.8%) together account for ~67% of Random Forest's predictive power, making billing behaviour the dominant signal in claim outcome prediction.

5. Logistic Regression severely underperforms (35.20% accuracy, Rejected F1=0.18), confirming the non-linear nature of claim outcome relationships.

5. Monitor Subgroup Fairness: Pune-based patients and SecureLife policyholders show slightly lower Rejected recall (68%). These subgroups should be monitored in production for systematic underprediction of rejected claims.